Inputs and specifications

>_ experiment.py
pre / execute
compare / audit
--config --output

VQE instance
Hamiltonian / circuit / grid
optimizer / shots / K / success
config/h2.json

Specification library
L1-L4 N1-N4 P1-P3
assumptions + runtime mappings
config/contracts.json

Shared execution model
M → A → V → C
phases / IDs / counters
controller + event context

Stage 1 | Pre-execution verification

closed_loop.pre_execution

Logical adapter
M → A → V → C
controller.smv
NuXmv order / epochs
nuxmv.log -> L1-L4

Numeric adapter
$A(x) \implies Q(x, f(x))$
normalize / sum / grid
Dafny real / integer
proof logs -> N1-N4

Probabilistic adapter
$\mathcal{N}(\mu, \sigma^2)$
Gaussian kernel -> DTMC (z,j)
Storm Python
$p_G(k)$ / $E[T \mid T \leq K]$

Verification record
pre_execution.json
configuration_digest
sources / artifacts / assumptions
compiled monitor + provenance

verifier outputs + model predictions

verify record / sources / config

Stage 2 | Instrumented VQE execution

execution / checker / temporal
faults: boundary / trace mutation

Candidates
z − +
current / - / +
candidate_indices

θ

Quantum kernel
q0 H R_z Y H R_z Y
q1 H R_z Y H R_z Y
state -> Born law
physics.point()

p

Group sampling
ZI/ZZ YZ XX
$[n_{00}, n_{01}, n_{10}, n_{11}]$
multinomial draws

n

Energy / score
$\widehat{E} = \sum_i c_i \hat{e}_i$
weighted Pauli terms
execution_values()

s

Optimizer / stop
$\arg\min \{s_0, s_-, s_+\}$ ties: 0,-,+
success / budget / abort
optimizer_commit / terminal

next index / coordinate

emit(kind, **data) measurement | evaluation | numeric_valid | optimizer_commit | terminal

continue / stop

online + persisted events

RuntimeChecker.consume(event)
Python online + replay: IDs / arithmetic / shots
DejaVu: temporal-input.tsv / dejavu.log

Trace store | *.jsonl.gz
run_id / evaluation_id
counts / shots / scores_used

Run summaries
independent seeded streams
success_step / physical_shots

Stage 1 record

Cross-stage comparison and deliverables

comparison.py + statistical reporting

Match + agreement
property + prospective scope
runtime check + assumptions
L1-L4 / N1-N4 verdicts

Probability + resources
$p_G(k)$ vs. $\hat{p}(k)$
P2/P3: conditional means
Wilson / bootstrap / Holm

Diagnostic results
requirement + event ID
assumptions / missing inputs
verdict vector / fault traces

Report artifacts
REPORT.md
verdict_agreement.json
probabilities.csv

A agreement D disagreement M assumption mismatch I inconclusive

manual revision + new verification record

Solid: module data / artifacts Orange dashed: events / replay Purple: control